CMSC 731 · Spring 2026

FRAMES *THAT* SPEAK

An augmented-reality museum treasure hunt where movement, close looking, and in-place interpretation are the learning mechanism.

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Motivation & Background

Why this matters — and why AR is not automatic.

The Attention Problem

Studies show most visitors spend only 15–30 seconds in front of an individual artwork. More time looking → more meaningful experiences.

Information Overload

Wall labels and audio guides deliver facts but pull attention away from the artwork itself. Visitors need support for knowing what to notice.

AR as a Candidate

AR could keep interpretation attached to the artwork: anchoring context, directing the eye. But it's only justified if it does more than deliver the same content through a novel interface.

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State of the Art & Gaps

What existing systems do — and what they miss.

Museum AR Overlays

A 2022 meta-analytic review of 51 museum learning studies found AR produced positive effects on achievement. Yet most systems superimpose supplementary materials stopping short of structured learning.

Engagement ≠ Learning

One art-museum study found AR increased visitors' liking of artworks more, while a book guide produced stronger learning gains. Engagement and learning do not automatically follow each other.

Living-Portrait Precedent

Snap's 2026 Jonathan Yeo exhibition and Louvre collaboration showed AR-powered portraits that respond in real time are viable, but neither is designed around a learning structure.

Gap: No system combines AR overlays + living portrait moments inside a theme-based learning path.

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Proposed Approach

Three workstreams, integrated into one experience.

AR Recognition & Anchor Pipeline

Unity + AR Foundation: image tracking per artwork, spatial anchoring of overlays, content delivery from recognition to in-scene interaction.

Unity · AR Foundation · ARKit / ARCore

Interaction Engine & Themed Hunt Logic

Clue progression, theme-based branching, prompt triggering, state management across artworks, and UI flow for treasure-hunt completion.

C# · State Machines · UI Toolkit

Generative Living Portrait Pipeline

Image-to-video workflow turning static paintings into animated portrait moments, integrated into AR via contextual triggers.

Image-to-video models · Python · Asset pipeline

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Challenges

Recognition

How does the system know which artwork is in front of the user?

Anchoring

How do overlays stay visually attached to a real painting as the user moves?

Integration challenges for both with Meta SDK

While Pre-existing drag and drop frameworks for both tasks exist on mobile devices, like ARKit and Vuforia, it is currently not possible to integrate them easily onto the quest platform

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Recognition and Anchoring approaches

QR Marker Recognition approach

We explored the QR anchoring approach where Meta Quest detects a physical QR code as a trackable anchor and read its payload like portrait:demo_01. That payload is mapped to a specific video/portrait prefab, which is spawned on the portrait.
We did not use QR anchoring because every portrait would need a visible QR code. Also, QR markers must be printed large enough and clearly visible for reliable tracking, making the approach less practical for natural museum displays.

ARToolKit and Scenery based mobile approach

An alternate mobile AR approach was also explored using ARToolKit and a Scenery-based workflow, where image recognition and overlay placement were tested on a phone-first AR stack before moving fully into the Quest pipeline. This helped validate the basic interaction pattern. However, due to issues with integration with the quest platform, we had to shift to a different approach.

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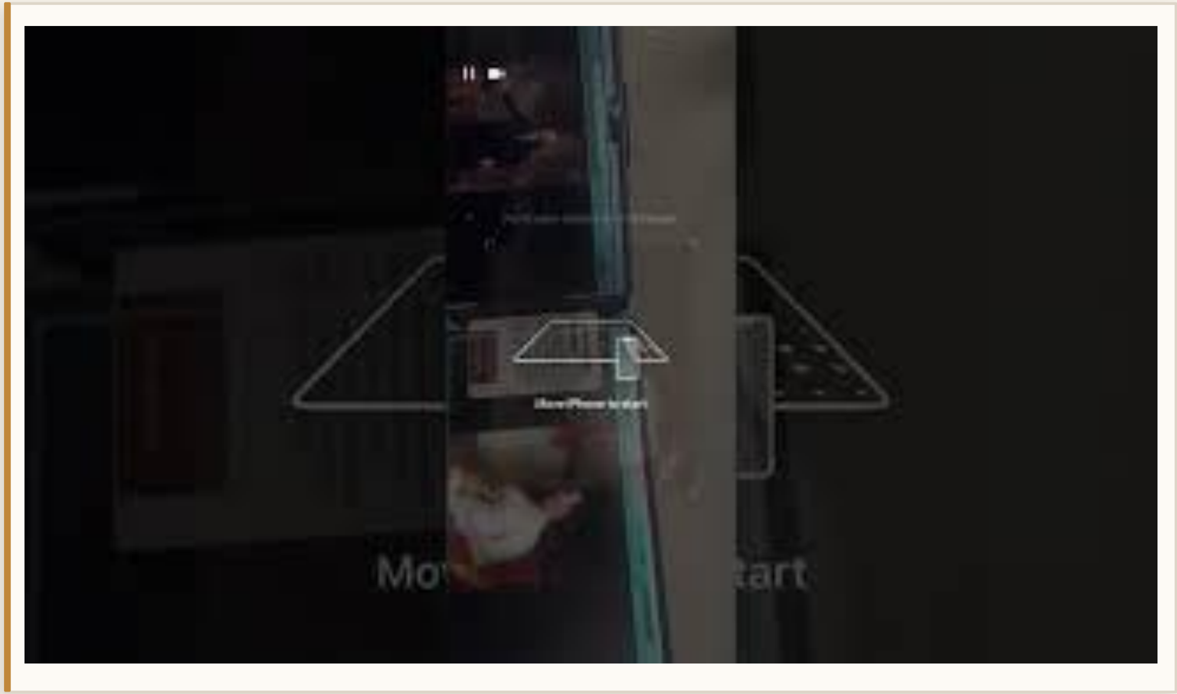
Recognition and Anchoring Phone Demo



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Recognition & Anchoring Implementation

Quest 3
Passthrough frame (camera API)

Live camera image from the headset

→

OpenCV ORB Match

Feature-match against target artwork

→

Camera & Env Raycast

Convert image position into a wall hit

→

Wall-Locked Overlay

Align video quad to detected surface

→

State Trigger

Recognition unlocks next interaction

Target Matching

One active artwork target is loaded at a time to ensure high performance.

Spatial Placement

Environment raycast anchors content directly to the physical wall surface.

Runtime Stability

Overlay freezes and detection pauses after lock to maintain stable alignment.

The system recognizes the artwork in the headset camera feed, converts the detection into a physical wall hit, then locks the living-portrait overlay to that surface.

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The user flow

Start Experience	Go to Artwork 1	Scan and Recognize	Reveal Clue / Animation	Respond & Solve	Next Step / Complete
Intro screen and hunt instructions	Visitor approaches the first selected frame	Headset detects the artwork and triggers the overlay	Living portrait and clue appear on the frame	Visitor answers or completes the prompt to continue	Progresses to the next frame until the final reveal

The experience moves the visitor through a linear sequence of scanning, revealing, and responding, with each artwork unlocking the next step in the hunt.

What Was Accomplished

IMPLEMENTED BY US

Quest 3 AR museum scene	*Arindam + Julian*
Artwork recognition + anchor stability	*Arindam*
Game Logic	*Julian*
Clue / overlay UI flow	*Shruti*
Living-portrait generation workflow	*Julian + Shruti*
Generative output → AR integration	*Shruti + Arindam*
Staged gallery + demo walkthrough	*All*

BORROWED COMPONENTS

Unity Engine	Core development platform
Meta SDK Building blocks	Camera Rig, Passthrough, Environment Raycast Manager
XR Interaction Components	Spatial interaction primitives
OpenCV	Image Recognition and anchoring
Image-to-video model (NeroAI)	Generative living-portrait backbone
Public artwork images (National Portrait Gallery open access Repository)	Inputs to the portrait pipeline

Course Integration

AR/MR Registration

Course concept: AR depends on real-time alignment between virtual content and the physical world.

In our project: The living portrait is registered onto the detected artwork/wall so interpretation appears spatially attached to the frame.

SLAM-Backed Spatial Tracking

Course concept: XR systems rely on headset tracking and spatial understanding to maintain stable virtual objects.

In our project: We use Quest’s built-in spatial tracking and environment raycasting for wall-locked overlays.

Perception-to-Rendering XR Pipeline

Course concept: XR systems combine sensing, tracking, rendering, and interaction in one real-time loop.

In our project: Passthrough, OpenCV ORB, and Unity combine to detect artwork and render animated content.

Attention-Aware AR Interaction

Course concept: XR interaction design should manage user attention, not just add interface elements.

In our project: The AR layer keeps interpretation in the same visual field as the art, reducing scan-to-learning friction.

Results & Demo

Quest 3 AR museum scene

Staged gallery with printed art reproductions and spatial overlays.

One complete hunt walkthrough

End-to-end: recognition → clue → overlay → living portrait → completion.

Generative portrait integration

Static paintings animated via image-to-video pipeline, triggered on recognition.

Live Demo: Staged Gallery Walkthrough

Live Website: Frames that Speak

Video Demo

Future Work

Robust Artwork Recognition

Replace manual selection with automated recognition across diverse lighting and viewing angles.

Multiple Gallery Themes

Support symbolism, patronage, composition, and historical context across the same collection.

Large User Study

Run a structured study measuring recall, attention duration, and learning outcomes across themed hunts.

Glasses-Based Deployment

Port to Snap Spectacles or similar hardware for hands-free, in-gallery use at scale.

References

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THANK *YOU!*